

HOMEWORK 3 (FP)

PROBLEM I – PERFECT RISK SHARING (IVÁN WERNING)

Consider a finite group I of individuals. Income for each individual $y_t^i(s_t)$ is determined each period as a function of the current state of nature¹ $s_t \in S$ (where S is a finite set). As in class, s^t denotes a history and $\Pr(s^t)$ is the probability of that history as of period 0. Denote aggregate income by $Y_t(s_t) = \sum_{i \in I} y_t^i(s_t)$. Let utility for individual i be given by $E \sum_{t=0}^{\infty} \beta^t (u^i(c_t))$ which under our assumption on uncertainty is

$$E \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \Pr(s^t) u(c_t(s^t))$$

1 – Assume that there is no aggregate savings technology, that the state of nature is observable and that there are no commitment problems.

Write out the Pareto problem for given Pareto weights $\{\lambda^i\}_{i \in I}$. Show that at the optimum consumption for individual i , $c_t^i(s^t)$, can be written as depending only on aggregate income in that period – once we control for $Y_t(s^t)$, consumption does not depend additionally on s^t .

2 – We now generalize the previous result. Assume there is a “storage technology”: if in period $t-1$ an amount $S_t(s^{t-1}) \geq 0$ was put aside for storage, then in period t an amount $(1+r_t(s_t))S_t(s^{t-1})$ is available (for consumption or storage) in addition to any current income $Y_t(s^t)$. Show that a similar result as in question (1) holds but that now we must condition on total consumption $C(s^t) = \sum_{i \in I} c_t^i(s^t)$. (note that we impose the non-negativity constraint on storage, thus our result in question 1 can be thought as a special case where $r_t = 0$ so that at the optimum $S_t = 0$ and thus $C_t = Y_t$).

3 – Let the utility function be of the CARA form:

$$u^i(c) = -\frac{1}{\gamma^i} \exp(-\gamma^i c)$$

Show that consumption takes the form: $c_t^i = a^i C_t + b^i$ where a^i and b^i are constants and $\sum a^i = 1$ and $\sum b^i = 0$. How does the distribution of γ^i affect a^i and b^i ? How does the distribution of Pareto weights λ^i affect a^i and b^i ?

4 – Let the utility function be of the CRRA form:

$$u^i = \frac{c^{1-\sigma}}{1-\sigma}$$

where the risk aversion σ is assumed to be the same for all individuals. Show that consumption takes the form $c_t^i = \alpha^i C_t$ with the constants α^i satisfying $\sum \alpha^i = 1$. How do the constants α^i depend on the Pareto weights λ^i ?

PROBLEM II – LUCAS’S CCAPM (FROM ECONOMETRICA, 1978)

Consider a world with a single representative agent in which a random and exogenous amount of perishable output d_t falls from a “fruit tree” each period. (There is no other output.) Output follows the stochastic process

$$\log d_t = \log d_{t-1} + \varepsilon_t, \quad E_{t-1} \varepsilon_t = 0 \tag{1}$$

where the i.i.d. shock ε_t is distributed $N(0; \sigma^2)$. There is no way to grow more fruit trees.

The agent’s lifetime utility function is:

$$E_t \sum_{s=t}^{\infty} e^{-\theta(s-t)} u(c_s)$$

¹Note that s_t summarizes the entire distribution of current income and possibly contains additional information, e.g. forecasts of future income.

where $\theta > 0$ is the rate of time preference. Assume there is a competitive stock market in which people can trade shares in the fruit tree, whose price on date t is denoted by p_t . (If you buy a share on date t , you get your first dividend only on date $t + 1$, so p_t is an “ex dividend” price.)

1 – Show that a representative individual will choose contingent consumption plans such that on each date,

$$p_t u'(c_t) = e^{-\theta} E_t [(p_{t+1} + d_{t+1}) u'(c_{t+1})]$$

(use the individual budget constraint that $c_t + p_t z_{t+1} \leq (d_t + p_t) z_t$, where z_t is the share of the fruit tree the individual holds at the end of period t)

2 – Show that in equilibrium, the fundamentals or bubble-free price of the tree is:

$$p_t^* = E_t \sum_{s=t+1}^{\infty} e^{-\theta(s-t)} \frac{u'(d_s)}{u'(d_t)} d_s$$

3 – Interpret this formula in terms of expected payoffs and a “risk premium”. What the sign of the risk premium on the tree, and why?

4 – Let agents have the utility function:

$$u(c) = \frac{c^{1-\rho} - 1}{1-\rho}$$

where $\rho > 0$. Show that equation (1) implies

$$E_t(d_s^{1-\rho}) = d_t^{1-\rho} e^{\sigma^2(1-\rho)^2(s-t)/2}$$

(To do this you will need to look up properties of the lognormal probability distribution in a statistics text.)

5 – Deduce from question 4 that if $\rho > \sigma^2(1-\rho)^2/2$ then $p_t^* = \chi d_t$ where

$$\chi = \frac{1}{e^{(\theta - \sigma^2(1-\rho)^2/2)} - 1}$$

6 – Now return to a general strictly concave utility function $u(c)$. Let b_t be the random variable $A d_t^\lambda u'(d_t)$, where $\lambda = \sqrt{2\theta/\sigma^2}$ and A is an arbitrary constant. Show that $p_t^* + b_t$ will satisfy the individual’s Euler equation (recall question 1) in equilibrium, so that b_t is an asset-price bubble. Show that $p_t = p_t^* + b_t$ violates the (equilibrium) transversality condition

$$\lim_{T \rightarrow \infty} e^{-\theta(T-t)} E_t [u'(d_{T+t}) p_{T+t}] = 0 \quad (2)$$

7 – Together with the equilibrium Euler equation,

$$p_t u'(c_t) = e^{-\theta} E_t [(p_{t+1} + d_{t+1}) u'(c_{t+1})] \quad (3)$$

condition (2) is sufficient for a stochastic price path $\{p_t\}$ to be an equilibrium. Here we will show the necessity of (2). Iterate (3) forward to derive

$$p_t u'(d_t) = E_t \left[\sum_{s=t+1}^{\infty} e^{-\theta(s-t)} u'(d_s) d_s \right] + \lim_{T \rightarrow \infty} e^{-\theta(T-t)} E_t [u'(d_{T+t}) p_{T+t}]$$

(This is the same step leading to the solution p_t^* in question 2. Argue that if the limit above is strictly negative, everyone would want to buy more of the tree and never sell it. Then argue symmetrically that if the limit is strictly positive, we cannot be looking at an equilibrium either. Why? Individuals could then raise expected lifetime utility through the following strategy: sell a tiny amount of the fruit tree today and consume the proceeds of that sale now, never repurchasing the portion of the fruit tree sold (that is, reduce the asset demand z_t , which equals 1 in the hypothesized equilibrium, permanently.) Why does the Euler equation (3) not suffice to rule out the possibility that such a strategy raises expected lifetime utility?